Nukavarapu Yaswanth Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.50
Sri Chaitanya	Class XII (Andhra Pradesh)	2019	9.88%
Sri Chaitanya	Class X (Andhra Pradesh)	2017	10.00%

SKILLS

Analytics: Python, SQL, Excel, Power BI | **Tools:** PowerPoint, Git

Data Analysis: Statistical Analysis, Exploratory Data Analysis, Data Visualization

Business: Financial Modelling, Market Analysis, Business Strategy, Process Improvement

Quantitative: Optimization, Statistical Modelling, Quantitative Analysis

PROJECTS

E-Commerce Customer Segmentation & LTV Marketing Strategy *(Git)*

Python, SQL, Excel, Statistics, Power BI

- Processed 150k+ purchase histories using **RFM modeling** and **K-Means Clustering** (evaluated via Silhouette Score of 0.68) to segment buyers into 5 behavioral personas.
- Formulated targeted **retention strategies** for high-churn segments, projecting a 15% increase in repeat purchases through decision-tree-based campaign modeling.

Dynamic Price Elasticity & Revenue Maximization Engine *(Git)*

Python, SQL, Statistics, Optimization

- Modeled price elasticity using **Log-Log Regression** and **Generalized Additive Models (GAMs)** across 15+ product categories using historical sales data.
- Applied constrained non-linear optimization (**SLSQP**) to generate optimal price points per segment, projecting a **6.8% margin lift** without losing volume.

B2B Enterprise Revenue & Cohort Retention Analytics *(Git)*

Python, SQL, Excel, Power BI, Statistics

- Analyzed 2+ years of SaaS transactional data using **SQL window functions** to perform cohort retention modeling and calculate Customer Lifetime Value (**CLV**).
- Built interactive **Power BI dashboards** tracking ARR, churn velocity, and expansion revenue, surfacing pricing tier gaps that unlocked an estimated 11% ARR growth.

Multi-Echelon Supply Chain & Network Optimization*(Git)*

Python, SQL, Excel, Optimization, Operations Research

- Formulated a Mixed-Integer Linear Programming (**MILP**) model in **PuLP** to optimize inventory allocation and route logistics across 4 distribution centers under strict lead-time constraints.
- Executed **Monte Carlo sensitivity analysis** on fuel spikes and demand surges, identifying network re-routes that cut total logistics costs by 14.5%.

Retail Demand Forecasting & Inventory Analytics *(Git)*

Python, Pandas, SQL, XGBoost, PyTorch, Power BI

- Built a daily forecasting pipeline using **PyTorch LSTM-XGBoost ensembling**, achieving an **8.4% MAPE** and reducing RMSE by 22% over baseline SARIMAX.
- Integrated seasonal Fourier features and automated SQL data pipelines into interactive **Power BI dashboards** to visualize stockout risks and regional demand trends.

WORK EXPERIENCE

Tata Consultancy Services (TCS), Pune : Machine Learning Engineer

- Developed **Association rules** among attributes with High viability using Association rule mining (**Apriori algorithm**)
- Developed a ML model (**Random Forest**) with **R^2 of 91%**, which predicts the cytotoxicity of nanomaterials
- Improved model by reducing attribute count using **Feature Selection** algorithms and **Visualized** Random Forest tree

PUBLICATION

Machine Learning-Based Cytotoxicity Prediction of Nanoparticles (ACS)

Machine Learning, Predictive Analytics

- Analyzed published nanoparticle cytotoxicity data to develop **predictive models** for quantitative and qualitative safety assessment.
- Identified key factors influencing cytotoxicity and demonstrated a more **robust classification** approach requiring fewer input attributes.

POSITIONS OF RESPONSIBILITY

Ramanuja Hostel, IIT Hyderabad - Hostel Secretary

- Coordinated **hostel affairs** and served as a liaison between residents and administration, facilitating resolution of **student concerns** and **day-to-day issues**.

EXTRACURRICULAR

Intellify × NSS -- Educational Support Volunteer

- **Taught** Chemistry to Class 11-12 students from underprivileged backgrounds through asynchronous sessions, supporting their preparation for public examinations.